

MAPLE PLAINS, PE, Canada

WMO: 710267

Lat: 46.300N

Lon: 63.580W

Elev: 150

StdP: 14.62

Time Zone: -4.00 (NAA)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.7	-1.6	-13.2	2.7	-4.9	-8.1	3.6	0.3	26.3	29.6	23.4	25.1	6.3	290	0.601

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.2	80.6	69.0	77.9	67.2	75.5	65.8	71.1	77.7	69.4	75.3	67.8	73.1	8.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.8	106.3	74.9	67.3	100.8	73.0	65.6	95.1	71.3	35.0	77.6	33.6	75.5	32.3	73.3	76.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
21.8	19.0	16.7	DB	-15.4	85.3	4.9	2.3	-19.0	87.0	-21.8	88.3	-24.6	89.6	-28.2	91.3	
			WB	-15.4	73.9	5.0	1.5	-19.0	75.0	-21.9	75.8	-24.7	76.7	-28.3	77.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.2	18.6	18.6	25.9	37.5	48.2	57.3	65.5	64.6	57.2	47.3	37.7	26.9
	DBStd	18.18	11.11	10.22	8.63	7.21	6.78	6.57	5.03	5.43	7.12	7.48	8.15	9.47
	HDD50	4368	975	878	747	380	118	11	0	0	17	145	379	718
	HDD65	8468	1440	1298	1212	824	521	241	56	75	248	550	820	1183
	CDD50	1534	0	0	0	6	62	231	481	452	233	60	9	0
	CDD65	161	0	0	0	0	0	11	72	62	15	1	0	0
	CDH74	856	0	0	0	0	10	68	385	338	54	1	0	0
	CDH80	103	0	0	0	0	0	5	49	42	7	0	0	0
Wind	WSAvg	7.8	8.8	8.6	8.9	8.5	7.9	7.1	6.4	6.0	6.7	7.6	8.4	9.1
Precipitation	PrecAvg	41.5	3.0	2.7	3.0	3.0	3.1	3.5	3.0	3.4	3.9	4.4	3.9	4.1
	PrecMax	55.6	5.0	5.5	5.5	5.6	7.7	5.9	5.6	6.3	8.1	8.0	6.7	9.1
	PrecMin	31.1	0.5	0.3	0.9	1.4	0.9	1.2	0.7	0.8	0.7	1.1	1.5	1.8
	PrecStd	5.7	1.1	1.1	1.2	1.1	1.5	1.2	1.2	1.6	2.0	2.0	1.3	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.4	46.0	51.7	63.7	75.3	79.9	84.7	84.2	79.3	70.8	62.4	53.2
		MCWB	49.1	44.7	44.3	54.8	60.2	65.8	71.4	70.2	69.2	62.8	58.8	51.8
	2%	DB	42.2	41.1	45.0	58.1	69.4	75.7	80.9	80.6	74.7	65.6	58.5	47.7
		MCWB	41.3	39.5	41.7	52.0	57.7	64.5	69.8	68.7	66.1	59.7	56.1	46.4
	5%	DB	37.9	36.7	40.8	53.3	64.5	72.6	78.2	77.7	71.4	62.4	54.8	43.5
		MCWB	36.4	34.8	37.8	48.6	54.1	62.6	68.0	67.3	64.1	57.7	52.3	42.3
	10%	DB	33.9	33.3	37.5	49.5	60.5	69.3	75.6	75.1	68.6	59.5	50.6	39.5
		MCWB	32.8	31.7	34.6	44.5	52.8	61.1	66.3	65.8	62.8	55.5	48.0	37.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.4	44.8	46.7	58.1	64.0	70.1	73.8	72.9	71.1	65.2	59.4	52.1
		MCDB	50.6	45.6	48.6	61.2	71.2	77.2	81.6	80.6	76.8	69.0	61.3	53.1
	2%	WB	41.8	40.0	42.5	53.1	59.7	67.0	71.6	70.8	68.2	61.3	56.2	46.6
		MCDB	42.3	41.2	44.7	56.6	65.7	72.6	78.7	77.2	72.2	64.0	58.0	47.7
	5%	WB	37.0	35.3	38.4	49.4	56.8	64.7	69.7	69.2	66.1	58.7	52.4	42.4
		MCDB	38.1	36.4	40.5	52.7	61.8	70.0	76.0	75.1	69.7	61.9	54.2	43.3
	10%	WB	33.1	32.1	34.7	45.3	54.0	62.1	68.0	67.5	64.0	55.8	48.3	38.3
		MCDB	34.0	33.3	37.2	49.1	59.8	68.1	73.2	73.2	67.7	58.8	50.6	39.3
Mean Daily Temperature Range	5% DB	MDBR	15.2	16.4	14.8	15.0	18.1	17.7	18.2	18.8	18.5	15.7	12.8	12.2
		MCDBR	19.4	18.0	17.9	21.6	25.4	22.5	22.0	21.7	19.8	18.4	16.6	16.1
		MCWBR	20.0	17.1	15.3	15.6	15.3	13.3	12.4	12.0	13.0	13.8	15.2	16.6
	5% WB	MCDBR	19.3	18.2	17.6	18.4	20.5	18.3	17.7	18.1	16.1	16.1	16.1	16.3
		MCWBR	19.8	17.8	16.1	15.6	14.1	12.0	10.9	10.8	11.7	13.4	15.7	16.8
Clear-Sky Solar Irradiance	taub	0.301	0.289	0.319	0.347	0.366	0.381	0.407	0.391	0.356	0.331	0.309	0.289	
	taud	2.350	2.335	2.347	2.401	2.408	2.382	2.346	2.383	2.479	2.529	2.518	2.474	
	Ebn at Noon	251	282	288	289	285	281	271	272	273	265	250	246	
	Edh at Noon	23	29	33	35	36	37	38	35	30	24	20	19	
All-Sky Solar Radiation	RadAvg	350	576	1002	1329	1611	1751	1783	1574	1208	748	390	268	
	RadStd	24	75	91	122	150	134	120	100	105	77	47	24	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.59	N/A	N/A	N/A	N/A	

Nomenclature: See separate page